Speed First RL for 3D Reconstruction and NBV

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by

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CERTIFICATE FROM THE SUPERVISOR

This is to certify that the thesis entitled ***Speed First RL for 3D Reconstruction and NBV*** submitted by ***Arpan Jain***, who has been registered for the award of BS in Data Science and Engineering degree of Indian Institute of Science Education and Research Bhopal is absolutely based upon his own work under the supervision of *Dr. Sundaram Muthu* of Department of Data Science and Engineering at Indian Institute of Science Education and Research Bhopal and that neither his/her thesis nor any part of the thesis has been submitted for any degree/diploma or any other academic award anywhere before.

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Abstract

Recent advances in Neural Radiance Fields (NeRF) and Gaussian Splatting have significantly improved 3D reconstruction quality, yet efficient data acquisition remains a critical bottleneck. Existing pipelines rely on passive or heuristic sampling, leading to redundant observations or incomplete coverage in complex, occluded environments.

We propose a unified active perception framework that integrates Deep Reinforcement Learning (DRL) with 2D Gaussian Splatting (2DGS) to jointly optimize keyframe selection and Next-Best-View (NBV) planning. Formulated as a Markov Decision Process (MDP), our agent learns a closed-loop policy for both sensory filtering and viewpoint control.

Central to our approach is a multi-objective reward function that combines geometric information gain, novelty bonus, and coverage improvement, while penalizing travel cost, computational overhead, and zero-gain actions. Additional efficiency and terminal coverage bonuses further guide the agent toward globally optimal reconstruction strategies.

By coupling RL-driven decision-making with the efficiency of 2DGS, our method reduces redundant data, improves coverage, and enables high-fidelity reconstruction in resource-constrained settings, paving the way for scalable real-world active vision systems.

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Chapter 1

Introduction

1.1 Motivation

Three-dimensional reconstruction is a core problem in computer vision, robotics, augmented reality, remote inspection, and digital-twin generation. A useful reconstruction pipeline must not only recover accurate scene geometry, but must also do so efficiently enough for real deployments. This requirement becomes especially important in mobile systems, where sensing time, motion budget, storage, and compute are all limited.

Recent advances in neural scene representations, especially Neural Radiance Fields and Gaussian Splatting, have significantly improved reconstruction fidelity [1, 2, 3]. However, data acquisition in many practical systems is still largely passive. A camera follows a predefined path, records a long sequence of frames, and the reconstruction backend processes the entire sequence afterward using classical reconstruction stages such as structure-from-motion and multi-view stereo [4, 5]. Although simple, this approach often wastes resources because many observations are redundant, while difficult regions remain under-observed.

1.2 Problem Statement

The central challenge addressed in this thesis is the following: how can a system decide, online, which observations are worth keeping and where the sensor should look next so that reconstruction quality improves as quickly as possible? This is commonly formulated as the Next-Best-View (NBV) problem [6, 7, 8]. In practice, NBV selection should not be treated independently of frame filtering, because every additional observation also increases storage, optimization time, and computational load.

This motivates a speed-first formulation. In this context, "speed-first" does

not mean sacrificing reconstruction quality; rather, it means maximizing geometric progress per unit sensing and computation. The policy should seek informative views, avoid repeated observations, and achieve strong global coverage with a compact subset of frames.

1.3 Thesis Focus

This report studies a reinforcement learning based framework for efficient 3D reconstruction and NBV planning. The proposed system combines two complementary ideas. First, deep reinforcement learning is used to learn a closed-loop policy for viewpoint control and keyframe selection. Second, a geometry-aware reconstruction backend based on 2D Gaussian Splatting is used to convert the selected observations into a high-quality scene model [8, 3].

The main premise is that view planning and frame selection should be optimized jointly rather than treated as separate modules. A jump-based policy can skip redundant intermediate frames in easy regions and sample more densely near occlusions, boundaries, and complex geometry. Such a strategy has the potential to reduce motion cost and reconstruction time while preserving or improving coverage [9].

1.4 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 summarizes the concepts required for active 3D reconstruction, including NBV planning, neural scene representations, and reinforcement learning. Chapter 3 identifies the limitations of current methods and motivates the proposed thesis direction. Chapter 4 states the research objectives of the work, and Chapter 5 presents the methodology that will be followed to design, train, and evaluate the proposed system.

Chapter 2

Background

2.1 3D Reconstruction Pipelines

Classical 3D reconstruction uses visual observations to recover camera pose and scene geometry. In RGB-only settings, structure-from-motion estimates camera poses and multi-view stereo densifies the scene [4, 5]. In RGB-D settings, depth measurements simplify geometric recovery, but the final result still depends strongly on viewpoint diversity and sampling density. In both cases, dense passive capture increases redundancy and often leaves persistent blind spots in cluttered or occluded environments.

2.2 Active Perception and NBV Planning

Active perception treats sensing as a sequential decision-making problem. Instead of collecting all possible observations, the system evaluates the current partial scene and selects the next camera pose that is expected to contribute the most new information. This process is generally referred to as Next-Best-View planning [6, 7].

Traditional NBV methods rely on hand-crafted criteria such as visibility, frontier coverage, expected surface gain, or travel cost [6, 7]. These methods are intuitive and can be effective in structured environments, but they are typically greedy. Because they optimize local gain, they may miss better long-horizon strategies in scenes with severe occlusion or disconnected informative regions. More recent learning-based methods attempt to address this limitation by learning reusable policies for active reconstruction [8].

2.3 Neural Scene Representations

Neural scene representations have changed the reconstruction landscape by providing models that can synthesize novel views and encode scene geometry. Neural Radiance Fields achieve high-quality novel-view synthesis, while more recent methods also use uncertainty or information-aware criteria for active view selection [1, 10]. However, repeated rendering queries can be computationally expensive for online planning.

Gaussian Splatting provides a faster alternative by representing a scene with explicit primitives that can be rendered efficiently [2]. In particular, 2D Gaussian Splatting improves geometric faithfulness by modeling scenes using oriented surface-aligned disks rather than purely volumetric blobs [3]. This makes 2D Gaussian Splatting attractive as a reconstruction backend when geometry accuracy and rendering speed are both important. Recent active reconstruction systems have also begun to explore this direction for high-fidelity scene capture [9].

The distinction between volumetric 3D Gaussians and surface-aligned 2D Gaussians is illustrated in Figure 2.1. Figure 2.2 further shows how a 2D Gaussian splat is defined in a local tangent frame and then projected into image space for rendering.

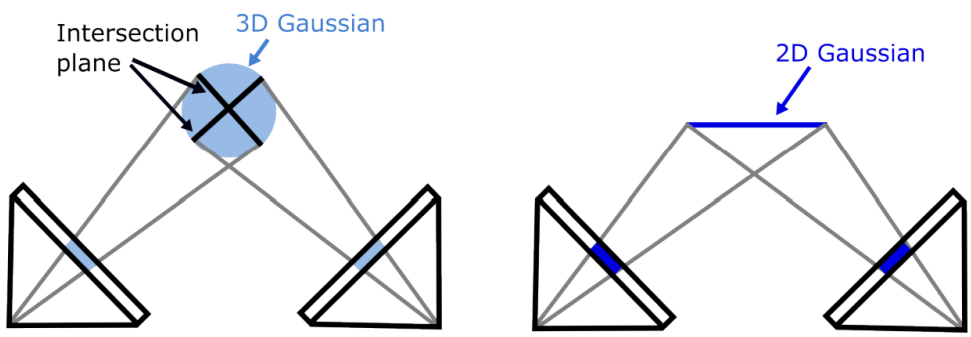

Figure 2.1: Conceptual comparison between volumetric 3D Gaussian primitives and surface-aligned 2D Gaussian primitives. The 2D formulation constrains the primitive to a tangent plane, which improves geometric faithfulness for reconstruction.

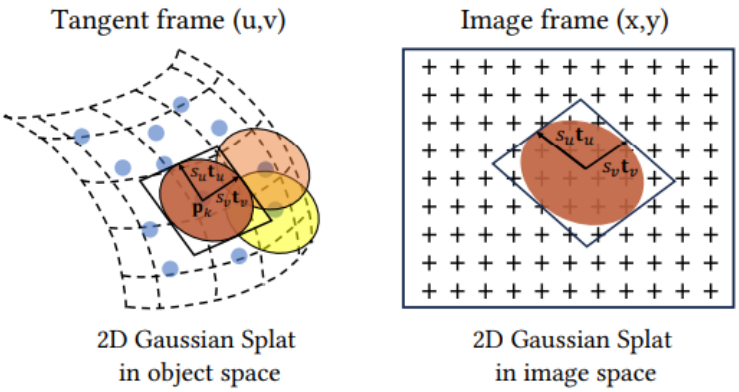

Figure 2.2: Illustration of a 2D Gaussian splat in object space and image space. The primitive is parameterized on a local tangent frame and then projected to the image plane during rendering.

2.4 Reinforcement Learning for View Selection

Reinforcement learning is well suited to NBV because the utility of an action depends on the current state of the partially observed scene and on future decisions that follow. An RL agent interacts with an environment, observes a state, chooses an action, receives a reward, and updates its policy to maximize long-term return. When formulated as a Markov Decision Process, active view selection can incorporate motion cost, reconstruction progress, coverage, and computational constraints within a single optimization framework [8].

Value-based methods such as Deep Q-Networks are particularly appealing when the action space is discrete. If actions are defined as frame jumps or candidate viewpoint indices, the agent can learn when to move conservatively and when to skip ahead. This allows the same policy to perform viewpoint planning and implicit keyframe filtering. In practice, variants such as Double DQN, dueling networks, and prioritized replay are often used to improve stability and sample efficiency [11, 12, 13].

2.5 Why This Combination Matters

The combination of reinforcement learning and 2D Gaussian Splatting is promising because the two components address complementary needs. Reinforcement learning provides adaptive, closed-loop control, while 2D Gaussian Splatting provides a fast, geometry-aware reconstruction backend. Together, they offer a practical path toward active reconstruction systems that are both accurate and efficient enough for real-world deployment [3, 9].

Chapter 3

Research Gaps

3.1 Limitations of Existing Approaches

Despite major progress in reconstruction and active perception, several important gaps remain.

1. **Passive acquisition remains inefficient.** Many reconstruction pipelines still assume dense capture along predefined trajectories. This leads to large numbers of near-duplicate frames, unnecessary storage and optimization cost, and poor use of sensing time [4, 5].

2. **Heuristic NBV strategies are often myopic.** Classical scoring rules usually optimize immediate gain, such as visible surface area or frontier expansion. In complex scenes, this can produce suboptimal trajectories that ignore long-term coverage and repeated occlusion patterns [6, 7].

3. **View planning and keyframe filtering are usually treated separately.** A system may first capture data and only later decide which frames are useful. This separation increases computational burden and prevents the planner from accounting for redundancy during acquisition itself [4, 9].

4. **Many learning-based NBV methods use coarse scene abstractions.** Several approaches reason over voxels, occupancy maps, or point statistics without directly coupling the policy to a geometry-aware reconstruction backend. As a result, the selected views may not be fully aligned with the needs of the final reconstruction model [8, 10].

5. **Speed and quality are not jointly optimized in a principled way.** In practice, a useful active reconstruction system must balance geometric gain against motion cost, backend latency, and zero-gain observations. Existing

methods often optimize only one or two of these aspects, which limits deployment on resource-constrained platforms [10, 9].

6. **Lightweight online decision making remains challenging.** High-fidelity representations improve reconstruction quality, but they are often too expensive to use directly inside the control loop. There is a need for a compact proxy that supports rapid decision making without disconnecting the planner from final reconstruction quality [1, 3].

3.2 Research Opportunity

These gaps suggest a clear opportunity for a unified framework that learns efficient NBV behavior while remaining tightly connected to a strong reconstruction backend. Such a framework should use a lightweight online representation for fast control, a reward function that explicitly captures the speed–accuracy trade-off, and a final reconstruction stage that benefits directly from the selected views [8, 3].

3.3 Problem Formulation for This Thesis

Motivated by the above gaps, this thesis asks the following research question: can a reinforcement learning policy select informative views and filter redundant frames quickly enough to improve 3D reconstruction efficiency without sacrificing scene coverage and geometric quality? The proposed work explores this question through a speed-first RL formulation coupled with a 2D Gaussian Splatting backend [8, 3].

Chapter 4

Objectives

4.1 Overall Objective

The overall objective of this thesis is to develop a speed-first reinforcement learning framework for active 3D reconstruction that jointly performs Next-Best-View planning and informative frame selection, and to couple this framework with a 2D Gaussian Splatting backend for accurate scene reconstruction.

4.2 Specific Objectives

1. To study the limitations of passive data acquisition and heuristic NBV planning for RGB-D based 3D reconstruction.
2. To design a compact state representation that captures partial scene coverage, camera motion, and online system load.
3. To formulate frame selection as a Markov Decision Process with a discrete jump-based action space so that keyframe filtering and viewpoint planning are addressed together.
4. To design a multi-objective reward function that balances geometric information gain, novelty, scene coverage, travel cost, and computational overhead.
5. To train and analyze a deep reinforcement learning agent capable of selecting informative observations from RGB-D sequences.
6. To integrate the selected observations with a 2D Gaussian Splatting backend for geometry-aware reconstruction.
7. To evaluate the proposed method against passive or heuristic sampling strategies using coverage, efficiency, and reconstruction quality metrics.

4.3 Expected Outcome

The expected outcome is a reconstruction pipeline that achieves better efficiency than dense passive acquisition by using fewer but more informative views. In addition to improved speed, the thesis aims to demonstrate that intelligent view selection can preserve or improve coverage in challenging scenes with occlusion and viewpoint redundancy.

Chapter 5

Methodology

5.1 Overview

The proposed methodology follows a two-stage design. An RL agent operates online on top of a lightweight geometry proxy to decide which frame or viewpoint should be selected next. The observations chosen by the policy are then forwarded to a 2D Gaussian Splatting (2DGS) backend that performs geometry-aware reconstruction. This separation allows fast control during acquisition while preserving high final reconstruction quality [8, 3].

Figure 5.1 summarizes the high-level pipeline from RGB-D input to RL-based selection and final 3D reconstruction.

Figure 5.1: High-level overview of the proposed speed-first reconstruction pipeline. RGB-D observations are encoded into state features, processed by the RL agent, and used for frame or Gaussian selection before final 3D reconstruction.

5.2 Problem Setup

Let the RGB-D sequence be represented as

$$\mathcal{D} = \{(I_k, D_k, T_{wc,k})\}_{k=0}^{N-1}, \tag{5.1}$$

where I_k is the RGB image, D_k is the depth map, and $T_{wc,k} \in SE(3)$ is the world-from-camera pose of the k-th frame. The agent starts from the first frame and iteratively selects the next frame to integrate into the reconstruction set.

For a pixel (u, v) with valid depth $z = D_k(v, u)$, the corresponding 3D point in the camera frame is computed using the camera intrinsics (f_x, f_y, c_x, c_y) as

$$\mathbf{p}_{cam} = \begin{bmatrix} (u - c_x)z/f_x \\ (v - c_y)z/f_y \\ z \end{bmatrix}. \tag{5.2}$$

The world-space point is then obtained as

$$\mathbf{p}_{world} = R_{wc}\mathbf{p}_{cam} + \mathbf{t}_{wc}, \tag{5.3}$$

where R_{wc} and $\mathbf{t}_{wc}$ are the rotation and translation components of $T_{wc,k}$.

5.3 Online Geometry Proxy

To support rapid decision making, the scene will be discretized into a voxel occupancy grid defined within fixed spatial bounds. Every selected depth observation contributes occupied voxels to the set $V_{obs,t}$ at time t. A reference set V_{gt} will be constructed from a sparse fusion of the full sequence so that progress can be measured consistently during training and evaluation. Lightweight occupancy summaries of this type are widely used to provide fast geometric context to active view-selection policies [8].

Coverage at time t will be estimated as

$$C_t = \frac{|V_{obs,t} \cap V_{gt}|}{|V_{gt}|}. \tag{5.4}$$

This proxy is lightweight enough for online control and provides a direct measure of how much of the scene has been explained by the selected observations.

5.4 Markov Decision Process Formulation

The active view-selection problem will be formulated as an episodic Markov Decision Process $\langle S, A, T, R \rangle$. Learning-based NBV policies are commonly expressed in this form because the utility of each sensing action depends on the current partial scene and its long-term effect on future reconstruction [8].

The state will combine three types of information: the current backend load,

the relative motion between consecutive selected views, and a pooled summary of the occupancy grid. A generic state representation may be written as

$$s_t = \left[q_t, v_t, \phi\left(T_{k_{t-1}}^{-1} T_{k_t}\right), \psi(G_t)\right], \tag{5.5}$$

where q_t denotes queue or backend load, v_t denotes motion-related features, $\phi(\cdot)$ encodes relative pose, and $\psi(G_t)$ summarizes the occupancy grid.

Instead of making a binary keep-or-drop decision, the action will be defined as a discrete frame jump

$$j_t \in \{j_{min}, \ldots, j_{max}\}, \tag{5.6}$$

which updates the selected frame index according to

$$k_{t+1} = \min(k_t + j_t, N - 1). \tag{5.7}$$

This jump-based design allows the policy to skip redundant frames in simple regions and sample more densely where the geometry is uncertain or occluded.

5.5 Reward Design

The reward used in the implemented NBV environment is more detailed than a single information-gain term. It combines geometric progress, motion cost, computation cost, coverage improvement, and trajectory regularization so that the agent is rewarded not only for seeing new parts of the scene, but also for doing so efficiently. Let n_t^{new} denote the number of newly observed voxels, d_t the travel distance, c_t the per-action compute cost, ΔC_t the change in coverage, C_t the total coverage at the end of an episode, and j_t the selected jump size. The step reward is defined as

$$\begin{aligned} R_t = w_1 \Delta I_{geo,t}^{sat} + N_t + B_t^{eff} - P_t^{travel} - P_t^{comp} \\ - P_t^{zero} - P_t^{jump} + B_t^{cov} + B_t^{term}. \end{aligned} \tag{5.8}$$

The geometric information term is intentionally saturated so that very large voxel-count spikes do not dominate the learning signal:

$$\Delta I_{geo,t}^{sat} = \left(\frac{n_t^{new}}{\alpha_{geo}}\right)^{\gamma}, \qquad N_t = w_n \Delta I_{geo,t}^{sat}, \tag{5.9}$$

where α_{geo} is a normalizing constant and $\gamma \in (0, 1]$ controls the compression strength. In the current implementation, $\alpha_{geo} = 50$ and $\gamma = 0.5$, which keeps the reward sensitive to improvement while suppressing extreme outliers.

Motion and computation are penalized in normalized form, while an efficiency bonus rewards views that reveal geometry cheaply:

$$\begin{aligned} P_t^{travel} &= w_{travel} \frac{d_t}{\alpha_{travel}}, \\ P_t^{comp} &= w_2 c_t, \\ B_t^{eff} &= w_{eff} \frac{\Delta I_{geo,t}^{sat}}{1 + d_t/\alpha_{travel} + c_t/\alpha_{comp}}. \end{aligned} \tag{5.10}$$

This efficiency term favors viewpoints that provide strong geometric gain without requiring large motion or expensive rendering/integration cost.

To discourage wasted actions and overly aggressive motion, the reward also includes a zero-gain penalty, an optional jump penalty, a step-wise coverage bonus, and a terminal coverage bonus:

$$\begin{aligned} P_t^{zero} &= \lambda_{zero} \mathbf{1}[n_t^{new} \leq 0], \\ P_t^{jump} &= \lambda_{jump} \rho(j_t), \\ B_t^{cov} &= \begin{cases} w_{\Delta C} \Delta C_t, & \Delta C_t > 0, \\ 0, & \text{otherwise}, \end{cases} \\ B_t^{term} &= \mathbf{1}_{term} w_T C_t, \end{aligned} \tag{5.11}$$

where $\rho(j_t) \in [0, 1]$ denotes the normalized jump size over the allowed jump range and $\mathbf{1}_{term}$ activates the terminal bonus only at the end of an episode. The jump penalty is disabled by default unless explicit trajectory smoothing is desired.

In the current reward implementation, representative default values are $w_1 = 1.0$, $w_n = 0.3$, $w_{travel} = 0.75$, $w_2 = 0.01$, $w_{eff} = 0.6$, $w_{\Delta C} = 10.0$, $w_T = 5.0$, $\alpha_{travel} = 0.10$, $\alpha_{comp} = 55.0$, and $\lambda_{zero} = 0.5$. The implementation also logs each reward component separately so that the contribution of geometric gain, efficiency shaping, penalties, and terminal bonuses can be analyzed during training.

This formulation explicitly encodes the speed-first philosophy of the thesis. The policy is encouraged to maximize reconstruction progress while avoiding unnecessary motion, repeated observations, excessive compute, and overly aggressive jumps. Reward designs that combine information gain, coverage, and efficiency are also consistent with prior active-view-selection work [6, 10].

5.6 Learning Algorithm

The policy will be trained using a value-based deep reinforcement learning algorithm. A Dueling Double DQN style architecture is suitable because it supports discrete actions, stabilizes value estimation, and can be combined with prioritized replay to improve sample efficiency [11, 12, 13].

During training, the agent will interact with RGB-D sequences episode by episode. At each step, it will select a jump action using an ϵ-greedy exploration strategy, observe the reward, store the transition in replay memory, and update the network parameters by minimizing temporal-difference error. A target network will be used to improve stability during optimization.

The training loop, including state construction, reward computation, replay-buffer updates, and target-network synchronization, is summarized in Figure 5.2.

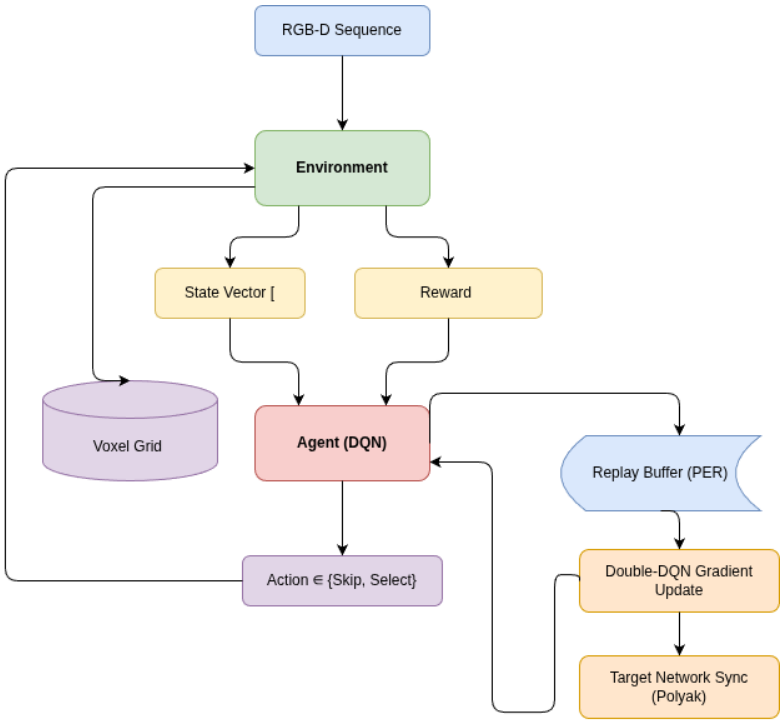

Figure 5.2: Detailed reinforcement learning loop used for frame skipping and selection. The environment produces the state vector and reward, the agent selects skip or keep actions, and learning is stabilized through prioritized replay and target-network synchronization.

5.7 Coupling with the 2DGS Backend

The observations selected by the learned policy will be exported to a 2D Gaussian Splatting backend. Each selected frame will contribute its image, depth,

intrinsics, and pose. The poses will be converted to the camera convention required by the backend, and an initial sparse point cloud will be generated from the chosen depth maps to stabilize optimization [3].

This stage ensures that the final reconstruction is not limited to the coarse occupancy representation used during planning. Instead, the occupancy grid guides fast decisions online, while 2DGS provides detailed geometry and rendering quality in the reconstruction stage.

5.8 Evaluation Plan

The proposed method will be evaluated against dense passive sampling and, where possible, heuristic frame-selection baselines on selected objects from OmniObject3D [14]. The evaluation will consider the following criteria:

- scene coverage achieved by the selected views,
- number of frames retained relative to the full sequence,
- travel distance or sensing effort,
- computational cost of the reconstruction pipeline, and
- qualitative and quantitative reconstruction quality from the 2DGS backend.

A successful outcome will show that the proposed speed-first RL policy achieves competitive or better reconstruction quality using a smaller, better curated set of observations.

Chapter 6

Results and Discussion

6.1 Experimental Setting

The experiments were conducted on a subset of the OmniObject3D dataset [14] containing the objects backpack_002, bed_001, glasses_001, keyboard_005, monitor_002, and suitcase_004. Three experimental settings were used in this study.

- **Exp A (compact skipping):** the RL policy was allowed to skip only 2 to 4 frames at a time.
- **Exp B (flexible skipping):** the RL policy was allowed to skip 3 to 8 frames, enabling more aggressive frame filtering when the scene was already sufficiently observed.
- **Exp C (no skipping):** a dense 2DGS baseline in which frames were rendered without skipping and all available observations were retained for reconstruction.

The reported values follow the metrics produced by the supplied evaluation pipeline in `main.py`. In particular, PSNR and SSIM are used to measure visual quality, coverage measures scene completeness, normalized Chamfer-L2 and normalized Chamfer-Sq measure geometric fidelity, and the number of selected frames, 2DGS finalize time, and peak GPU memory are used to evaluate efficiency.

6.2 Training Diagnostics

Figures 6.1 and 6.2 summarize the training and evaluation diagnostics for the two RL policies that perform frame skipping. Each dashboard reports episode reward, final geometry coverage, photometric quality (PSNR and SSIM), and

the jump sizes selected during evaluation. These diagnostics are useful because they show whether each policy learns the intended skip behavior before the final 2DGS reconstruction results are compared.

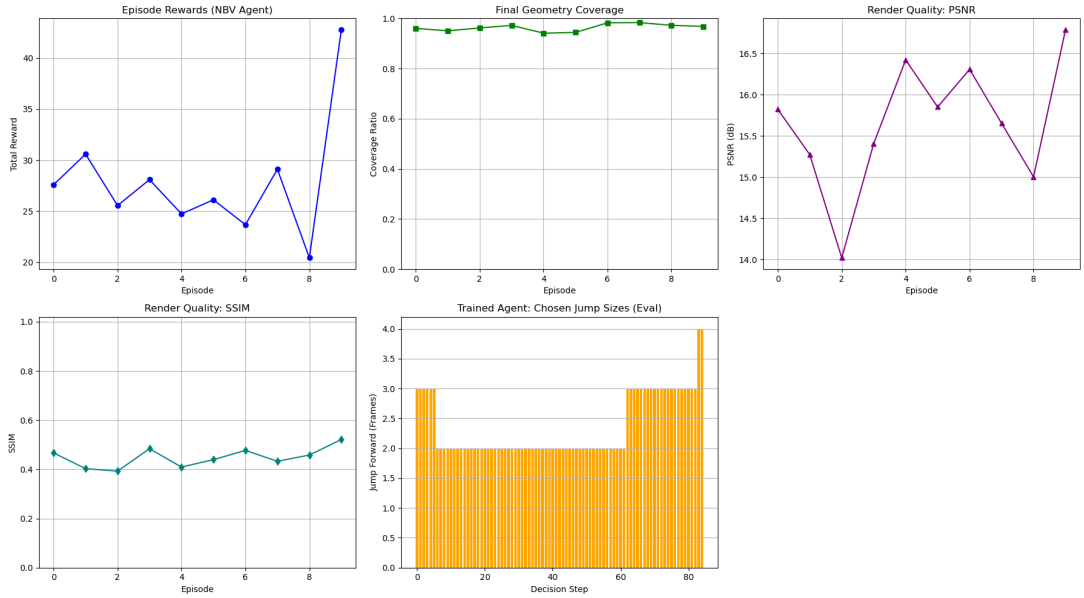

Figure 6.1: Training and evaluation diagnostics for Exp A. The compact policy keeps geometry coverage consistently high and mostly selects 2–3 frame jumps, with only occasional 4-frame skips during evaluation.

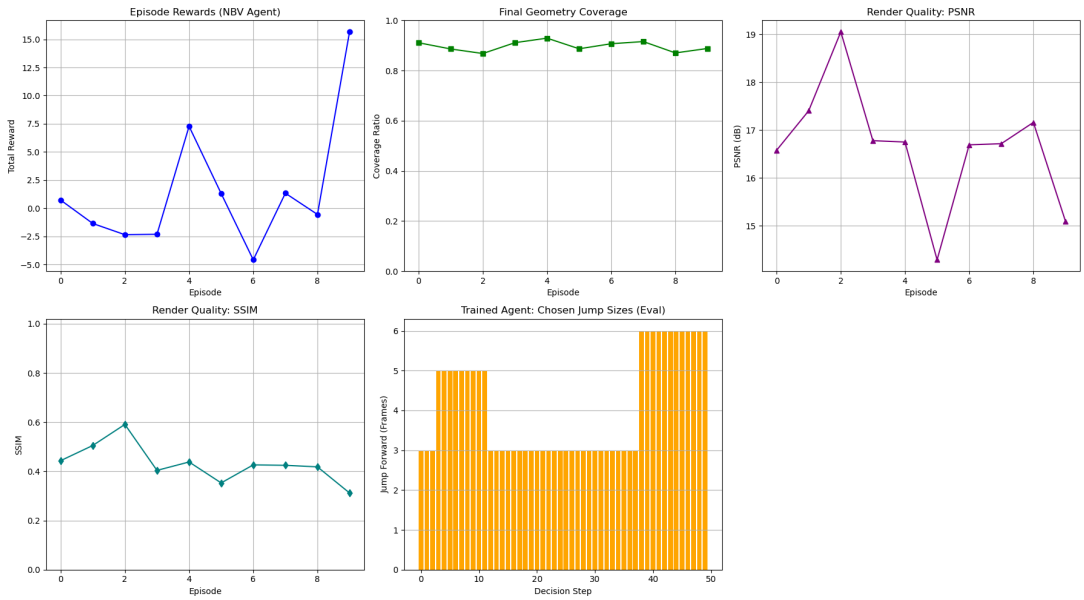

Figure 6.2: Training and evaluation diagnostics for Exp B. The flexible policy explores a wider jump pattern, producing more volatile reward and image-quality curves while maintaining high geometry coverage across episodes.

The compact Exp A policy shows steadier behavior: geometry coverage stays

in the high range across episodes, SSIM varies within a relatively narrow band, and the learned evaluation trajectory is dominated by 2- and 3-frame jumps. By contrast, Exp B exhibits a more aggressive decision pattern, with frequent 3- to 6-frame jumps and a more irregular reward curve. This higher variance is expected because the broader action space makes the control problem harder, but the policy still reaches strong coverage and achieves higher PSNR peaks in several episodes. Overall, the diagnostic plots indicate that Exp A behaves as a stable conservative skipper, whereas Exp B learns a more assertive compression strategy that is later reflected in its stronger speed–quality trade-off.

6.3 Quantitative Results

The uploaded metric matrix now provides complete entries for all three experiments on the full six-object evaluation set: backpack_002, bed_001, glasses_001, keyboard_005, monitor_002, and suitcase_004. Therefore, the aggregate comparisons in Tables 6.1 and 6.2 are reported over this full selected subset, enabling a direct comparison across Exp A, Exp B, and Exp C.

Table 6.1: Average quality metrics on the selected six-object OmniObject3D subset.

Experiment	**PSNR ↑**	**SSIM ↑**	**Coverage ↑**	**Norm. Chamfer-L2 ↓**	**Norm. Chamfer-Sq ↓**
Exp A (2–4 skip)	10.904	0.440	0.860	0.160	0.050
Exp B (3–8 skip)	**11.394**	**0.455**	0.746	0.161	0.050
Exp C (no skip)	10.237	0.423	**1.000**	**0.158**	**0.048**

Figures 6.3 and 6.4 provide a graphical view of the aggregate metrics reported in Tables 6.1 and 6.2 over the full six-object evaluation set. These plots make the trade-off between reconstruction quality and computational efficiency easier to compare across the three experimental settings.

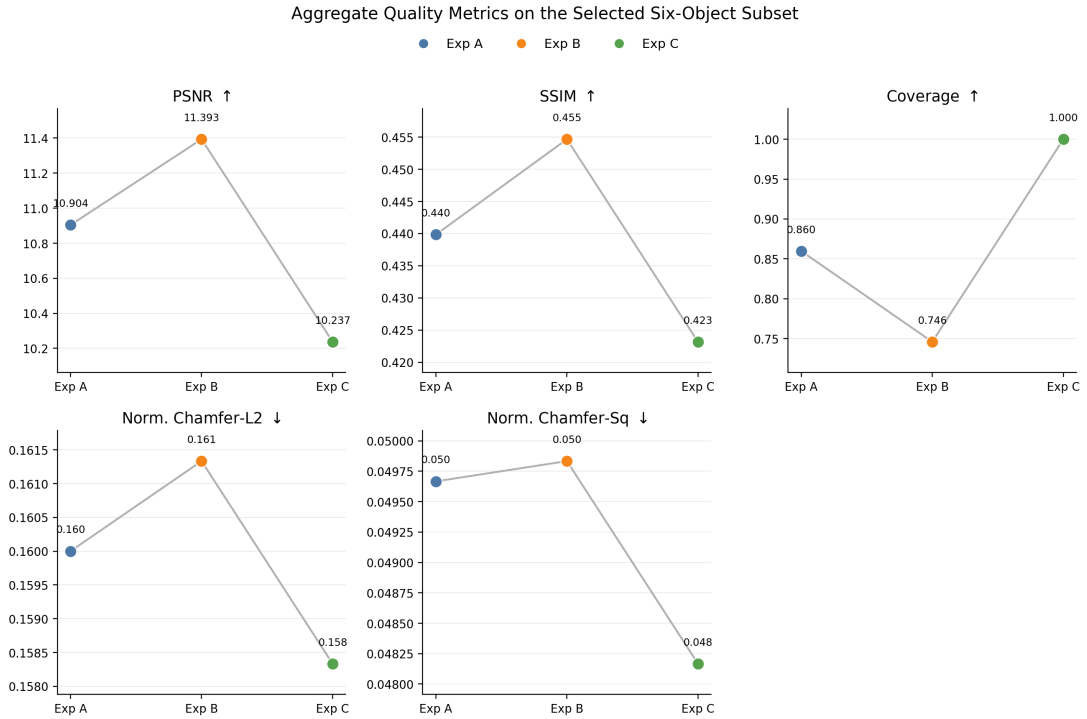

Figure 6.3: Graphical comparison of the average quality metrics on the selected six-object subset. Exp B achieves the strongest PSNR and SSIM, while Exp C remains strongest in coverage and geometric error.

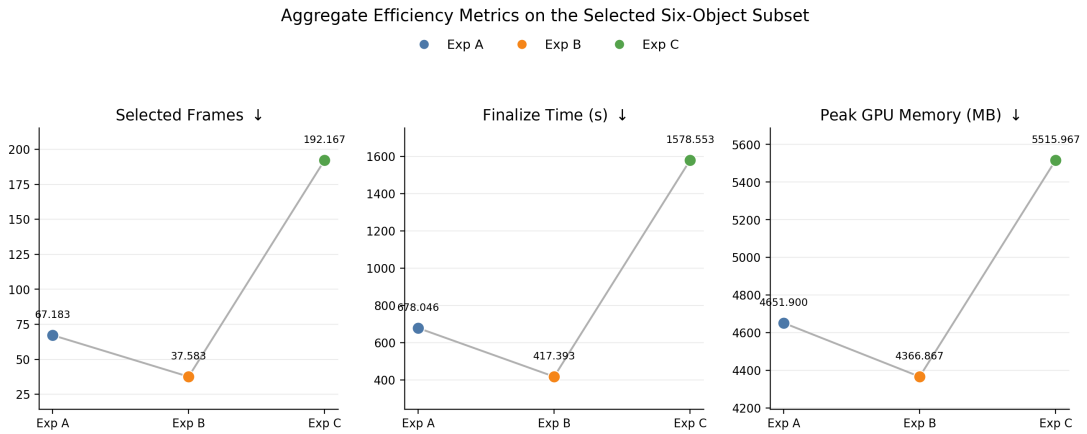

Figure 6.4: Graphical comparison of the average efficiency metrics on the selected six-object subset. Exp B consistently uses fewer frames, less finalization time, and less GPU memory than the other settings.

Table 6.2: Average 2DGS efficiency metrics on the selected six-object OmniObject3D subset.

Experiment	Selected Frames ↓	Finalize Time (s) ↓	Peak GPU Memory (MB) ↓
Exp A (2–4 skip)	67.183	678.046	4651.900
Exp B (3–8 skip)	**37.583**	**417.393**	**4366.867**
Exp C (no skip)	192.167	1578.553	5515.967

In addition to the separate aggregate quality and efficiency summaries above, Figures 6.5 and 6.6 directly compare computational cost against coverage and photometric quality. These views make it easier to see which experiment occupies the most favorable operating point when the resource budget is fixed.

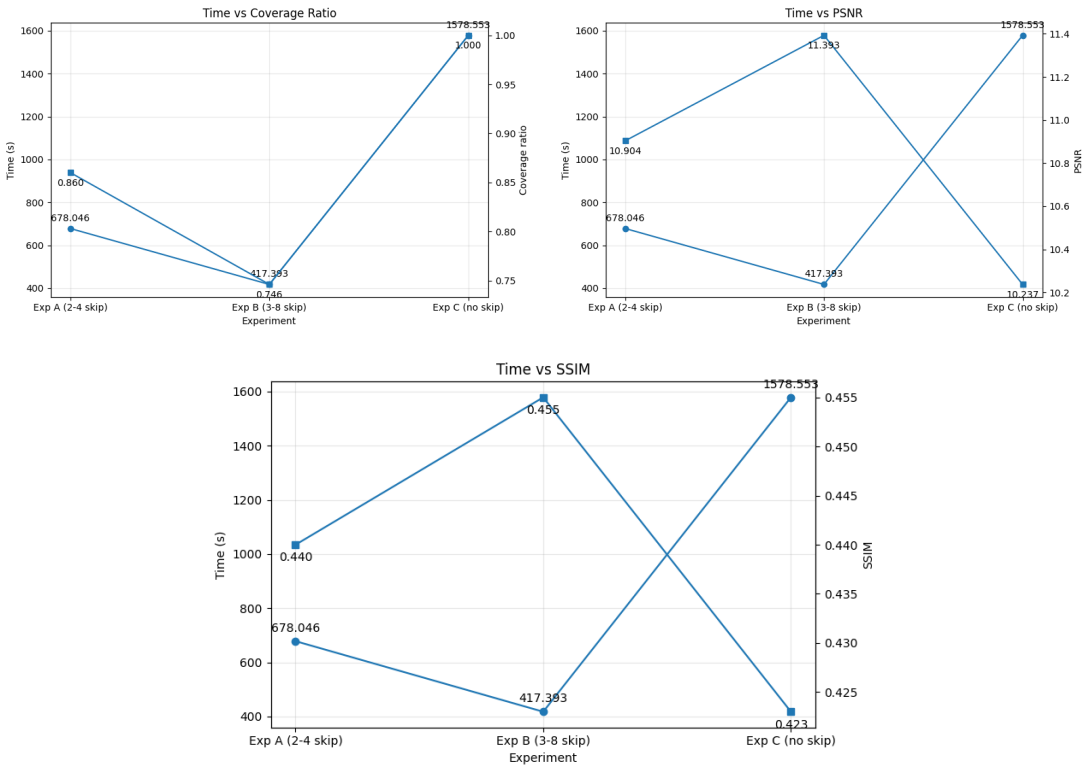

Figure 6.5: Pairwise trade-off plots between 2DGS finalization time and the average reconstruction metrics. Exp B achieves the lowest finalization time while also delivering the highest average PSNR and SSIM, whereas Exp C reaches full coverage only at much higher computational cost.

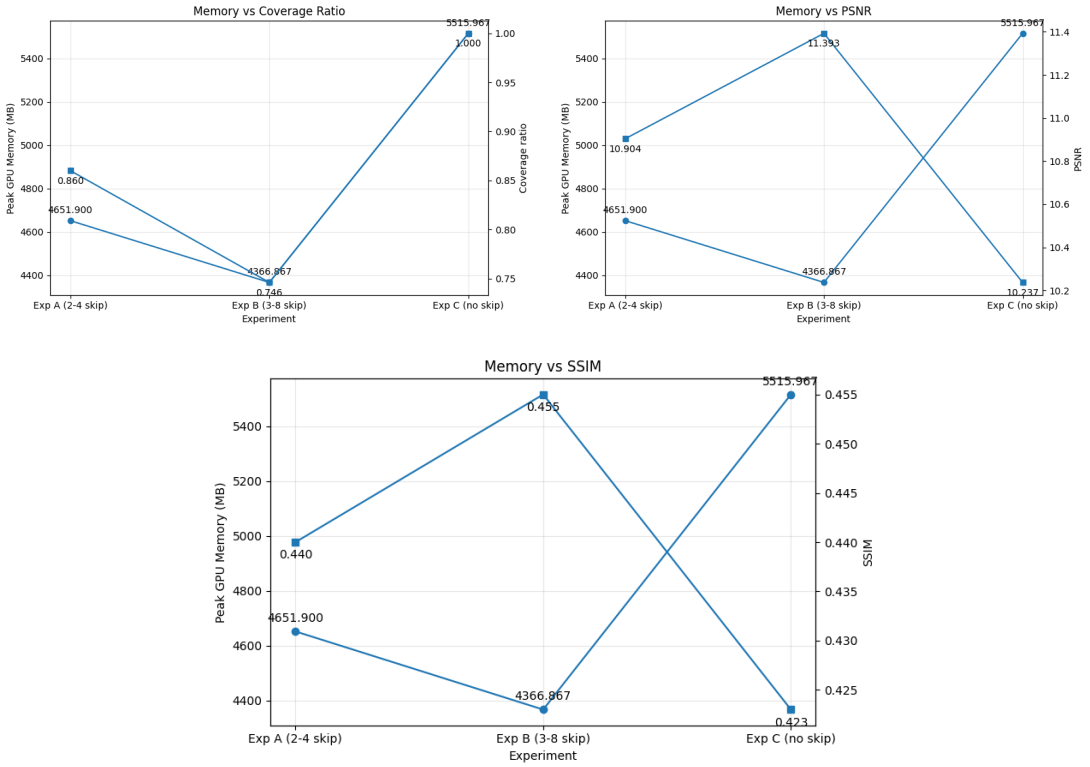

Figure 6.6: Pairwise trade-off plots between peak GPU memory and the average reconstruction metrics. Exp B uses the least memory while preserving the strongest photometric quality, whereas Exp C trades substantially higher memory for maximum scene completeness.

The pairwise plots reinforce the main conclusion of the chapter. When time or peak memory is treated as the constrained resource, Exp B occupies the most favorable region for PSNR and SSIM, making it the best practical operating point for speed-first deployment. Exp C remains the preferred choice only when absolute completeness is the primary target, since it is the only setting that reaches full coverage on average. Exp A stays between these extremes and is therefore useful as a conservative compromise when some additional coverage is desired without paying the full dense-processing cost.

6.3.1 Object-wise Metrics

Tables below report the per-object results for all three experiments. Bold values indicate the best result for a given metric within that object.

Table 6.3: Object-wise metrics for backpack_002.

Experiment	PSNR ↑	SSIM ↑	Coverage ↑	Norm. Chamfer-L2 ↓	Frames ↓	Finalize Time (s) ↓	Peak GPU (MB) ↓
Exp A	15.653	0.448	0.964	**0.127**	70.600	1054.779	5451.900
Exp B	**16.650**	0.431	0.897	0.129	**39.300**	**668.404**	**4881.200**
Exp C	14.501	**0.469**	**1.000**	0.128	201.000	2533.637	6478.100

Table 6.4: Object-wise metrics for bed_001.

Experiment	PSNR ↑	SSIM ↑	Coverage ↑	Norm. Chamfer-L2 ↓	Frames ↓	Finalize Time (s) ↓	Peak GPU (MB) ↓
Exp A	9.272	0.514	0.852	0.134	52.500	1141.749	4767.700
Exp B	**9.312**	**0.524**	0.736	0.148	**28.600**	**700.361**	**4700.400**
Exp C	9.136	0.508	**1.000**	**0.120**	148.000	2749.500	5702.500

Table 6.5: Object-wise metrics for glasses_001.

Experiment	PSNR ↑	SSIM ↑	Coverage ↑	Norm. Chamfer-L2 ↓	Frames ↓	Finalize Time (s) ↓	Peak GPU (MB) ↓
Exp A	7.298	0.129	0.845	**0.188**	68.400	917.198	5718.000
Exp B	**7.544**	**0.163**	0.676	0.192	**39.200**	**545.477**	**5370.500**
Exp C	6.503	0.123	**1.000**	0.199	201.000	2134.718	6866.400

Table 6.6: Object-wise metrics for keyboard_005.

Experiment	PSNR ↑	SSIM ↑	Coverage ↑	Norm. Chamfer-L2 ↓	Frames ↓	Finalize Time (s) ↓	Peak GPU (MB) ↓
Exp A	9.359	0.399	0.766	0.106	71.200	681.137	5978.600
Exp B	**11.159**	**0.486**	0.628	0.101	**39.100**	**382.476**	**5696.200**
Exp C	8.943	0.373	**1.000**	**0.099**	201.000	1536.394	7213.200

Table 6.7: Object-wise metrics for monitor_002.

Experiment	PSNR ↑	SSIM ↑	Coverage ↑	Norm. Chamfer-L2 ↓	Frames ↓	Finalize Time (s) ↓	Peak GPU (MB) ↓
Exp A	**15.055**	**0.817**	0.937	0.232	70.800	95.283	3220.000
Exp B	14.699	0.790	0.886	0.235	**38.600**	**69.255**	**3001.600**
Exp C	13.787	0.771	**1.000**	**0.224**	201.000	165.337	3507.600

Table 6.8: Object-wise metrics for suitcase_004.

Experiment	PSNR ↑	SSIM ↑	Coverage ↑	Norm. Chamfer-L2 ↓	Frames ↓	Finalize Time (s) ↓	Peak GPU (MB) ↓
Exp A	8.787	0.332	0.794	0.173	69.600	178.131	2775.200
Exp B	**8.997**	**0.334**	0.652	**0.163**	**40.700**	**138.385**	**2551.300**
Exp C	8.553	0.295	**1.000**	0.180	201.000	351.730	3328.000

Qualitative Exp B Frame-Selection Examples

Figures 6.7 and 6.8 show representative selected-view grids produced by Exp B for backpack_002 and keyboard_005. In both cases, the agent chooses frames from clearly different viewpoints instead of repeatedly keeping nearly identical observations. The backpack example includes front, side, and oblique views around the object body, while the keyboard example samples multiple orientations along the elongated structure. This indicates that the reward function is working well: distinct views are getting selected because the policy is being rewarded for novel and informative observations, which helps explain why Exp B can use fewer frames while still maintaining strong reconstruction quality.

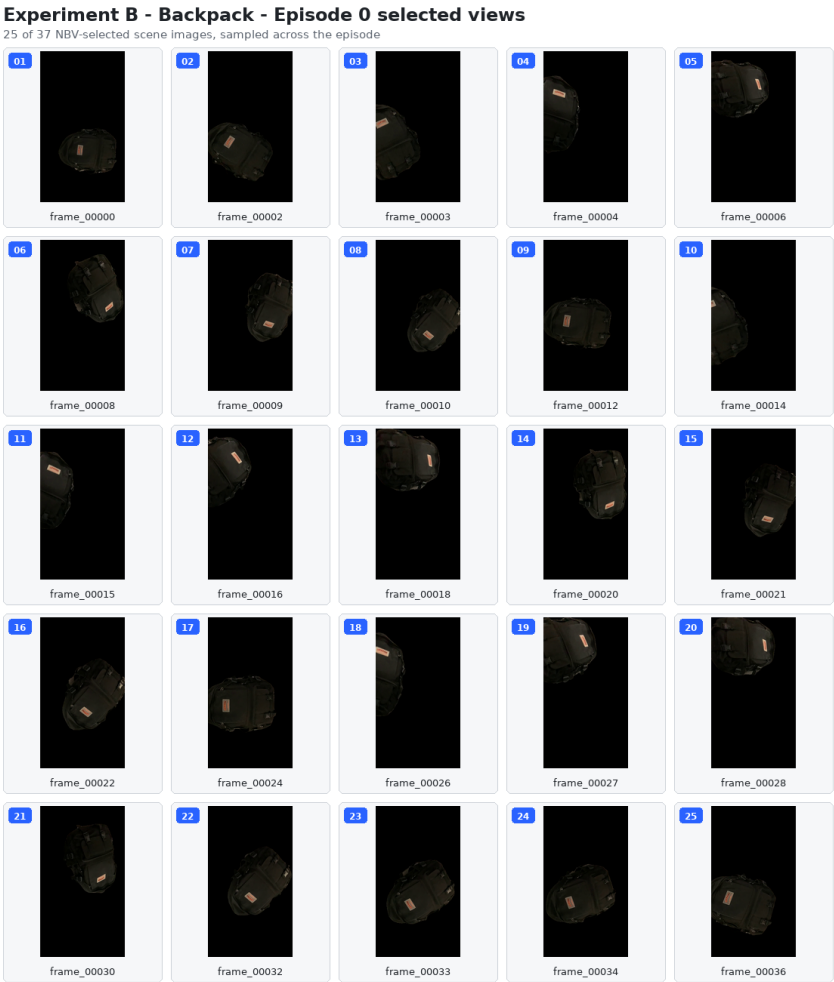

Figure 6.7: Selected views retained by Exp B for backpack_002. The kept frames cover clearly different object orientations, showing that redundant neighboring views are not being repeatedly selected.

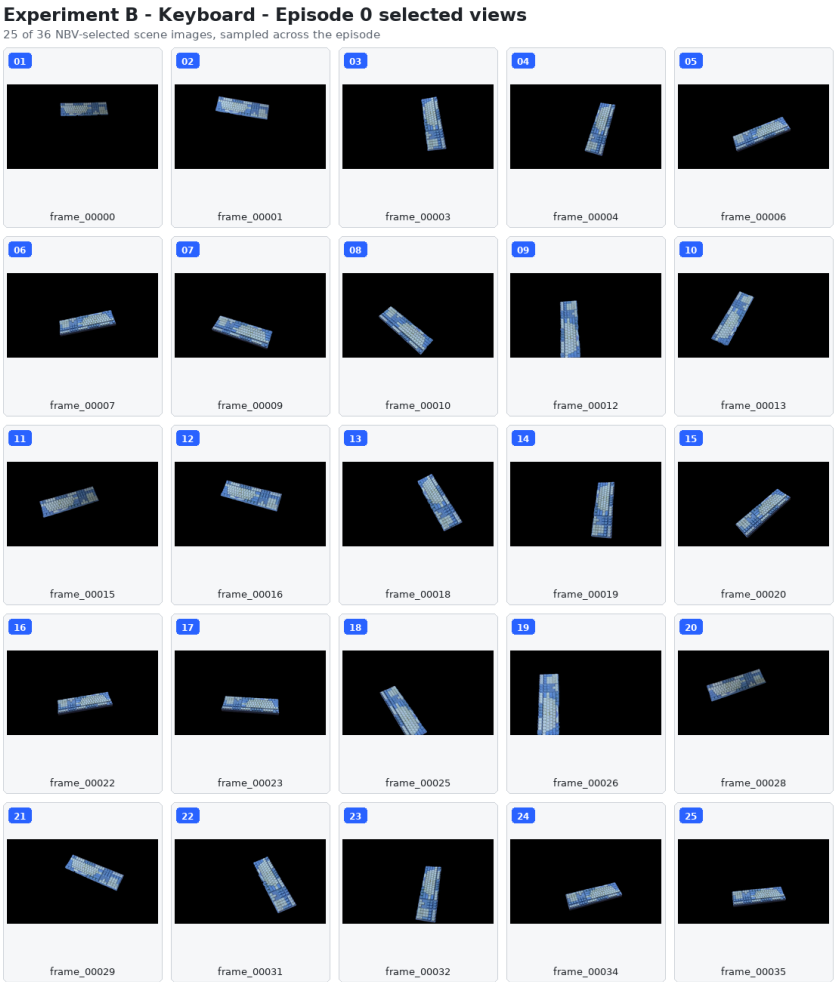

Figure 6.8: Selected views retained by Exp B for keyboard_005. Distinct viewpoints are again selected across the episode, suggesting that the reward function is effectively favoring novelty and informative coverage.

Figures 6.9, 6.10, and 6.11 summarize the object-wise trends across all quantitative metrics on the six-object evaluation set. For completeness, Figure 6.10 includes the object-wise normalized Chamfer-Sq values from the uploaded metric matrix, even though that metric is omitted from the per-object tables for readability.

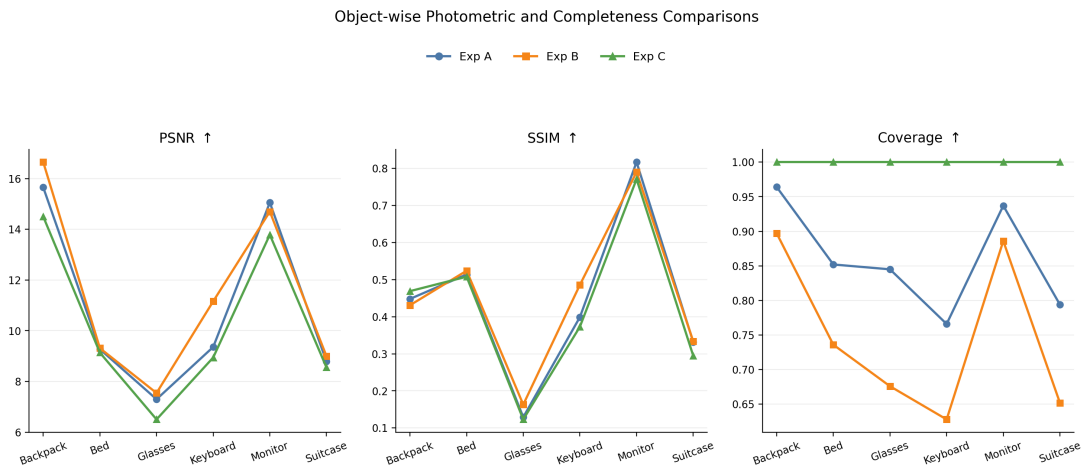

Figure 6.9: Object-wise comparison of PSNR, SSIM, and coverage across all six objects. Exp B usually leads in PSNR and SSIM, while Exp C dominates coverage across the full evaluation set.

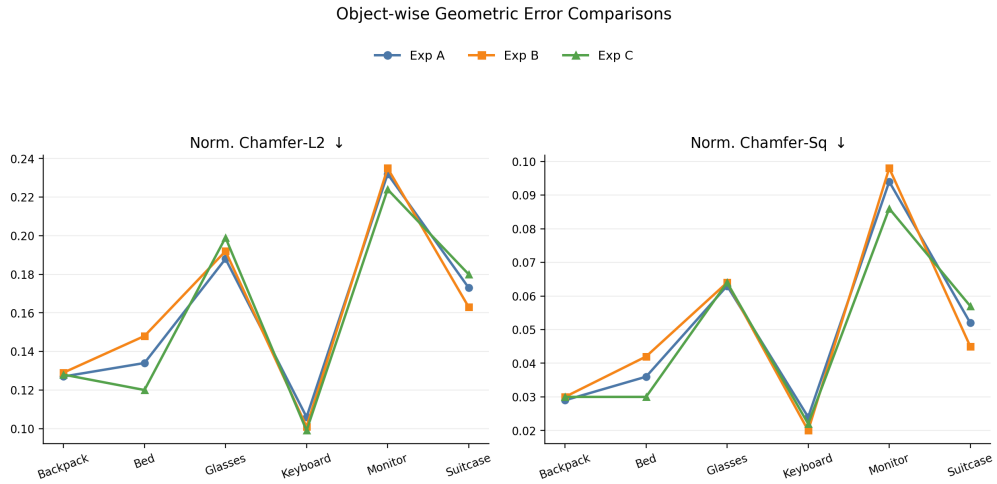

Figure 6.10: Object-wise comparison of the normalized Chamfer-L2 and normalized Chamfer-Sq metrics. These plots show that geometric accuracy remains competitive across the three settings, with dense reconstruction often favoring completeness and lower error when available.

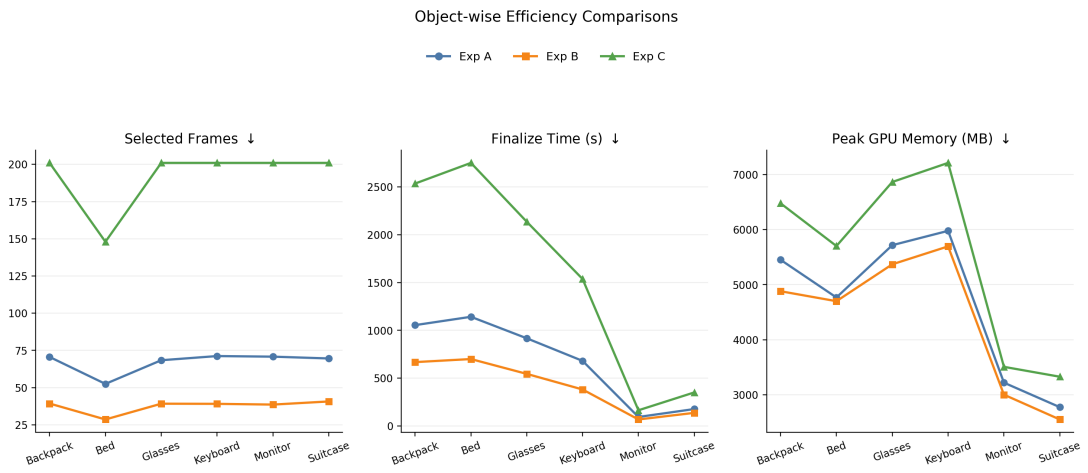

Figure 6.11: Object-wise comparison of selected frames, 2DGS finalization time, and peak GPU memory. Across all six objects, Exp B is the most efficient setting, while Exp C is consistently the most expensive.

6.4 Observations and Explanation

1. **Exp B provides the strongest speed–quality trade-off.** It achieves the best average PSNR and SSIM on the full six-object evaluation set while also selecting the fewest frames. Compared with Exp C, Exp B reduces the number of selected frames by about 80.4%, reduces 2DGS finalize time by about 73.6%, and lowers peak GPU memory by about 20.8%. This suggests that the wider 3–8 skipping range removes redundant views effectively without significantly hurting reconstruction quality.

2. **Exp C gives the best completeness, but at the highest cost.** The dense no-skipping baseline reaches perfect average coverage across all six objects and also produces the lowest average normalized Chamfer values. This indicates that retaining every available frame helps the system observe more of the object and improves geometric consistency. However, these gains come at a very high computational cost, since Exp C uses far more frames and requires the most time and GPU memory during 2DGS finalization.

3. **Exp A behaves as a conservative middle ground.** The compact 2–4 skip setting preserves better coverage than Exp B (0.860 versus 0.746) and slightly better normalized Chamfer-L2 (0.160 versus 0.161), but it is less efficient and produces lower PSNR and SSIM. This indicates that smaller allowable skips avoid overly aggressive filtering, yet they do not reduce computational load as strongly as Exp B.

4. **Object-wise trends support the aggregate tables.** Across all six objects, Exp B achieves the highest PSNR on five out of six objects and the highest SSIM on four out of six objects. Exp C achieves perfect coverage on all six objects and the lowest normalized Chamfer-L2 on three objects, while Exp A remains strongest on monitor_002 for PSNR and SSIM. Exp B is also the most efficient setting for selected frames, finalization time, and peak GPU memory on every object. Overall, these trends show that flexible RL-based skipping is most suitable when the goal is to reduce cost while preserving photometric quality, whereas dense processing is more appropriate when maximum completeness is the primary objective.

6.5 Discussion

The results support the main thesis hypothesis that frame skipping should not be treated only as data reduction; rather, it can be used as an active control mechanism for efficient 3D reconstruction. A compact policy such as Exp A is safer in terms of completeness, but the more flexible policy in Exp B gives a better operating point for practical use because it reduces the workload passed to the 2DGS backend while still retaining strong visual quality. In contrast, Exp C confirms that dense rendering remains a useful upper bound for coverage, but it is substantially more expensive and therefore less suitable for speed-first deployment.

Chapter 7

Conclusion

7.1 Summary of the Report

This report explored a speed-first reinforcement learning framework for active 3D reconstruction and Next-Best-View planning. The central idea of the work is that frame selection and viewpoint planning should be optimized jointly so that the reconstruction backend receives informative observations rather than a dense sequence containing substantial redundancy. To support this goal, the report motivated the use of reinforcement learning for sequential decision making, described a compact state and reward formulation for active frame skipping, and coupled the learned policy with a 2D Gaussian Splatting backend for geometry-aware reconstruction [8, 3].

7.2 Main Findings

The experimental study on selected OmniObject3D objects [14] shows a clear trade-off between completeness and efficiency. Among the tested settings, Exp B, which allows flexible skipping in the range of 3 to 8 frames, provides the strongest overall speed–quality trade-off. It achieves the best average PSNR and SSIM across the full six-object evaluation set while also using the fewest selected frames, the lowest 2DGS finalization time, and the lowest peak GPU memory among the three experimental settings.

Exp C, the no-skipping dense baseline, produces the highest coverage and the best average normalized Chamfer values on the full six-object subset. This confirms that dense observation remains a strong upper bound for geometric completeness. However, the gain in completeness comes at a substantial computational cost because the dense baseline processes far more frames and places a significantly higher burden on the 2DGS backend.

Exp A acts as a conservative middle ground. By restricting skipping to 2 to 4 frames, it retains better coverage than Exp B and remains competitive for geometric accuracy on some objects, but it does not match Exp B in photometric quality or computational efficiency. Taken together, these findings indicate that moderate to flexible frame skipping is not merely a compression mechanism; it can serve as an active control strategy that reduces redundant sensing while preserving useful reconstruction quality.

7.3 Concluding Remarks

Overall, the report supports the thesis hypothesis that reinforcement learning can be used to guide efficient observation selection for 3D reconstruction. The results suggest that a well-designed skipping policy can substantially reduce the workload passed to the 2DGS backend without severely degrading the final reconstruction, and in some settings can even improve photometric quality by focusing processing on more informative views.

Although the present study is limited to a selected subset of objects, the overall pattern is consistent and encouraging. These observations provide a strong foundation for the remaining stages of the thesis, including broader evaluation, further reward tuning, and the extension of the framework to more challenging scenes and real-world active capture settings.

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